

CSA1304-THEORYOF COMPUTATION LAB EXPERIMENTS

10. Write a C program to find ϵ -closure for all the states in a Non-Deterministic Finite Automata (NFA) with ϵ -moves.

AIM

Write a C program to find the ϵ -closure of all the states in a Non-Deterministic Finite Automata (NFA) with ϵ -moves.

CODE

```
#include <stdio.h>

#define MAX 10

int n;
int epsilon[MAX][MAX];
int visited[MAX];

void eclosure(int state)
{
    int i;

    if(visited[state])
        return;

    visited[state] = 1;

    printf("q%d ", state);

    for(i = 0; i < n; i++)
    {
        if(epsilon[state][i] == 1)
            eclosure(i);
    }
}

int main()
{
    int i, j;
```

```

printf("Enter number of states: ");
scanf("%d", &n);

printf("Enter epsilon transition matrix:\n");

for(i = 0; i < n; i++)
{
    for(j = 0; j < n; j++)
    {
        scanf("%d", &epsilon[i][j]);
    }
}

printf("\nEpsilon Closures:\n");

for(i = 0; i < n; i++)
{
    for(j = 0; j < n; j++)
        visited[j] = 0;

    printf("ε-closure(q%d) = { ", i);

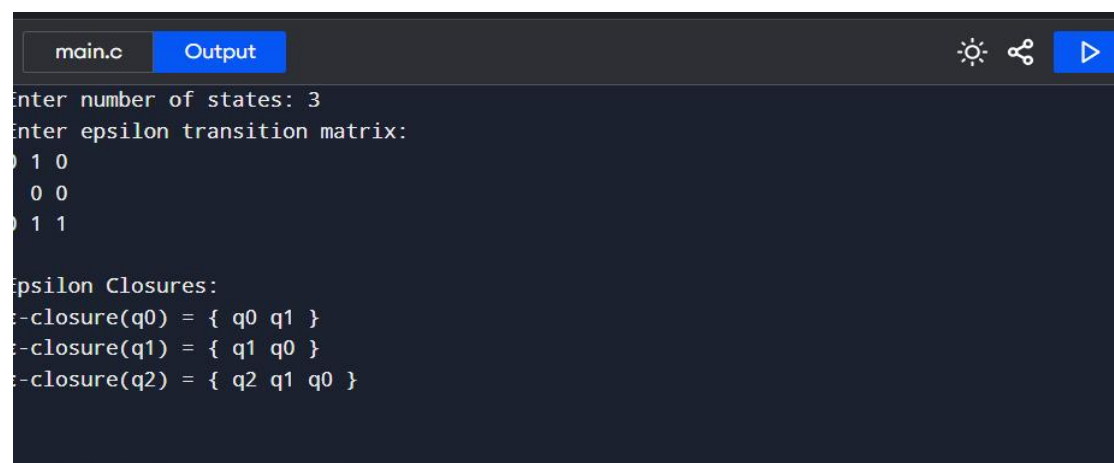
    eclosure(i);

    printf("}\n");
}

return 0;
}

```

OUTPUT



```

main.c  Output
Enter number of states: 3
Enter epsilon transition matrix:
0 1 0
0 0
0 1 1

Epsilon Closures:
ε-closure(q0) = { q0 q1 }
ε-closure(q1) = { q1 q0 }
ε-closure(q2) = { q2 q1 q0 }
Code Execution Successful

```